CodeBlue Workbook Architecture Guide

What each table is for, why it exists, and how the structure supports both the pitch and the technical build

One-sentence pitch: CodeBlue is not a single prediction model. It is a layered hospital outbreak decision-support architecture that separates evidence, pathogen logic, local deployment context, and reviewable policy actions so the system can adapt without becoming clinically confusing.

1. How to read the workbook

The workbook is easiest to understand if you read it as four connected layers rather than as one large spreadsheet. This is important for both the pitch and the technical implementation: it explains why the design is modular, auditable, and safer than a monolithic model.

- Evidence layer: literature-derived facts are split by causal stage so the model does not mix acquisition, severity, secondary infection, staff-vector logic, and outbreak confirmation.
- Pathogen pack: influenza-specific logic sits in a compact pack that captures timing, prioritized risk features, and intervention hooks.
- Deployment layer: local seasonality and current epidemiology are editable hospital inputs, not hardcoded pathogen biology.
- Policy layer: sources, triggers, actions, and trigger-action mapping convert evidence into reviewable operational decisions rather than automatic orders.

2. Core construction principles

Causal separation: The workbook was intentionally split so that factors known before infection stay separate from downstream severity or complication outcomes. This reduces leakage and makes the model easier to defend technically.

Human review: Actions are framed as reviewable decisions, not autonomous commands. This matters for medication, staffing, room assignment, and outbreak escalation.

Universal pathogen logic + local deployment inputs: Influenza biology is kept universal in the pathogen pack, while seasonality and current activity are externalized into deployment tables so a hospital can reconfigure local behavior without rewriting the pathogen logic.

Stable catalog + adaptive mapping: The action library is intentionally static. Adaptation happens in the trigger-action map, where seasonality, vaccination, ward vulnerability, spread state, and capacity modify what should happen.

Auditability: The policy layer starts from explicit source documents, then defines triggers, then defines actions, then defines the mapping between them. That chain is much easier to explain to judges, clinicians, and engineers.

3. Table-by-table summary

The workbook was constructed in layers. Grouping the summary by layer makes the logic easier to explain than presenting one very large undifferentiated table.

3.1 Evidence and feature-normalization sheets

Table	Layer	Rows	Purpose	Construction logic
ha_influenza_risk_factors	Evidence	91	Acquisition/protective factors for HAI.	Built from what is knowable before infection or early in hospitalization; downstream severity is excluded.
hospitalized_influenza_severity	Evidence	33	Severity after influenza is already present.	Used only after influenza is established, for ICU, mortality, LOS, and burden.
secondary_infections_in_influen	Evidence	12	Secondary infection after influenza.	Separated so post-influenza complications do not contaminate acquisition logic.
pathogen_interactions	Evidence	12	Which additional pathogens appear in overlap or superinfection.	Microbiology-oriented context, not a predictor sheet.
transmission_context_support	Transmission	48	Hospital spread mechanics: room, unit, cluster, ventilation, and controls.	Built from outbreak, rooming, exposure, and environmental studies.
staff_vector_transmission	Transmission	20	How staff act as exposure sources or vectors.	Built from HCW vaccination, presenteeism, asymptomatic infection, and contact studies.
advanced_outbreak_monitoring_to	Outbreak analytics	9	Sequencing and molecular outbreak clarification.	Reserved for WGS / sequencing once ordinary epidemiology is not enough.
library	Canonical feature	35	Canonical variable dictionary and alias normalization.	Stabilizes naming across extraction so features can be reused downstream.

3.2 Pathogen pack and deployment sheets

Table	Layer	Rows	Purpose	Construction logic
influenza_pack_metadata	Pathogen pack	-	Identity, scope, and governance of the influenza pack.	Universal pack object; local seasonality is intentionally external.
influenza_pack_timing	Pathogen pack	9	Universal influenza timing rules.	Built from CDC biological and healthcare-setting guidance.
influenza_pack_risk_features	Pathogen pack	14	Prioritized influenza risk/vulnerability features.	Compact prioritization layer over canonical features, not a duplicate of raw evidence.
influenza_pack_interventions	Pathogen pack	14	Influenza-specific intervention hooks.	Built from CDC infection-control and antiviral guidance; reviewable rather than auto-prescriptive.
deployment_seasonality_profile	Deployment	1	Editable local seasonality profile for a hospital geography.	Lets hospitals change alert windows without rewriting pathogen biology.

3.3 Policy and reference sheets

Table	Layer	Rows	Purpose	Construction logic
policy_source_library	Policy	8	Authoritative sources that justify triggers and actions.	Built from CDC and official public-health sources so the policy layer is auditable.
policy_trigger_library	Policy	21	Operational events that should make CodeBlue pay attention.	Triggers are event definitions only; they are not actions.
action_library	Policy	19	Stable catalog of reviewable actions.	Reusable action set; direct inputs stay here, while adaptation is pushed into the mapping layer.
trigger_action_map	Policy	34	Adaptive decision layer linking triggers to actions.	Encodes seasonality, vaccination, vulnerability, spread state, and capacity modifiers.
abreviações	Reference	39	Glossary of abbreviations used in the workbook.	Reference aid for human readers, not part of the model logic itself.

*Row counts are approximate active records in the reviewed workbook version, excluding template padding rows.

4. What each layer does in plain language

4.1 Evidence layer

The evidence sheets are the scientific memory of the system. They store extracted findings from literature in a consistent structure. The most important design choice is that they are split by causal stage, because that prevents the model from treating consequences of influenza as if they were pre-infection predictors.

ha_influenza_risk_factors: This sheet answers: who gets hospital-acquired influenza, or what reduces that risk? It is the acquisition sheet and is the main place for baseline vulnerability, early clinical signals, structural exposure, ward exposure, and protective factors that are knowable at or before the prediction time.

hospitalized_influenza_severity: This sheet starts only after influenza is already present. It answers: given established influenza, what predicts ICU use, mortality, longer stay, or worse outcomes?

secondary_infections_in_influen: This sheet is for what happens after influenza when bacterial, fungal, or resistant secondary infection develops. It exists so post-influenza complications do not pollute the acquisition logic.

pathogen_interactions: This is the microbiology context sheet. It does not ask who gets influenza; it asks which additional organisms appear when overlap or superinfection occurs.

transmission_context_support: This is the hospital-spread mechanics sheet: same-room exposure, same-unit exposure, ventilation, control bundles, masking, clustering, and environmental context.

staff_vector_transmission: This isolates staff-source evidence: vaccination coverage, presenteeism, asymptomatic infection, and staff-role contact patterns.

advanced_outbreak_monitoring_to: This sheet is intentionally narrow. It covers sequencing and molecular outbreak clarification once ordinary epidemiology is no longer enough.

4.2 Canonical feature layer

The library sheet is the normalization layer. It turns multiple literature expressions into canonical feature names, stages, and classes. This matters technically because it prevents the system from learning several near-duplicate versions of the same variable simply because papers described them differently.

4.3 Pathogen pack

The pathogen pack is not another literature dump. It is the compact influenza-specific logic that sits on top of the evidence layer.

influenza_pack_metadata: Defines the pack itself: identity, scope, target setting, governance, and the rule that seasonality is external rather than hardcoded.

influenza_pack_timing: Captures universal influenza timing rules: incubation, contagious window, default precaution duration, and prolonged-shedding modifiers.

influenza_pack_risk_features: Prioritizes the most operationally important influenza-related vulnerability and protection features. It is a weighting/prioritization layer, not the full evidence base.

influenza_pack_interventions: Defines the influenza-specific intervention hooks the system can route into the policy layer: screening, work restriction, droplet precautions, rooming, AGP precautions, antiviral review, and so on.

The key design choice here is universality: influenza biology stays in the pathogen pack, while geography-specific seasonality is stored elsewhere. That makes the pack portable across hospitals and regions.

4.4 Deployment layer

The deployment layer exists because hospitals do not all run under the same local respiratory-virus calendar. In the reviewed workbook, deployment_seasonality_profile is the editable local profile. This means the hospital can change pre-alert windows, high-alert windows, campaign dates, and local operating assumptions without changing the pathogen pack.

For the pitch, this is powerful because it shows the project is not just 'influenza in Ceará'; it is a reusable platform instantiated for Ceará. For the technical audience, it shows a clean boundary between universal pathogen logic and local configuration.

4.5 Policy layer

The policy layer is the operational brain. It converts evidence and pathogen logic into reviewable decisions.

policy_source_library: Stores the external authorities that justify the operational logic. It is the traceability layer.

policy_trigger_library: Defines events that should make CodeBlue pay attention. Triggers are not actions; they are operational conditions or signals.

action_library: Defines the stable catalog of reviewable actions. It does not hold the full adaptive decision logic; it holds what action exists, who owns it, what inputs it needs, and how it should be audited.

trigger_action_map: This is the adaptive rules layer. It connects triggers to actions and adds modifiers such as seasonality, vaccination context, vulnerability, spread state, and capacity pressure.

5. Why the structure is good for the pitch

For judges or non-technical reviewers, the main message is that CodeBlue is not trying to solve everything with one black-box score. Instead, it separates four questions that hospitals actually face:

Pitch message	Workbook answer
What do we know from evidence?	The evidence layer stores structured findings without mixing causal stages.
How does influenza behave?	The pathogen pack stores influenza-specific timing, vulnerability, and intervention logic.
How does this hospital differ locally?	The deployment layer stores editable local seasonality and current activity.
What should the hospital reviewably do?	The policy layer converts signals into triggered, auditable, human-reviewed actions.

That makes the pitch stronger because it shows the project is a hospital decision-support architecture, not only a literature review and not only a machine-learning model.

6. Why the structure is good for the technical build

For the technical collaborator, the structure solves common implementation problems early:

- It reduces feature leakage by separating acquisition from severity and post-influenza complications.
- It avoids unmaintainable logic by keeping the action catalog stable and moving adaptation into the trigger-action mapping.
- It makes localization cleaner by externalizing seasonality instead of hardcoding geography into pathogen biology.
- It keeps the path from source → trigger → action → mapping auditable and testable.
- It leaves room for later additions such as threshold configuration, hospital deployment profile, override logs, or pathogen-pack variants without redesigning the evidence base.

7. Recommended verbal explanation for the demo

A simple way to explain the workbook live is:

1. First, these literature tables are our evidence memory, but they are deliberately split so we do not mix acquisition, severity, secondary infection, staff-vector logic, and outbreak confirmation.
2. Second, we compress influenza-specific logic into a pathogen pack: timing, prioritized risk features, and intervention hooks.
3. Third, local seasonality is not hardcoded into influenza itself. It is an editable deployment input, which makes the architecture transferable across hospitals and geographies.
4. Fourth, the policy layer converts signals into reviewable actions. The system does not auto-order medication or irreversible actions; it proposes, prioritizes, and routes them for human review.

8. Practical notes for the tech handoff

- If a model is built later, train and validate from the evidence/canonical-feature side, not directly from the policy tables.

- Treat policy_trigger_library as event definitions and action_library as verbs. Do not re-collapse them into one table.
- Use trigger_action_map as the place for adaptive prioritization rules, because that is where seasonality, vaccination, vulnerability, spread state, and capacity can be expressed cleanly.
- Keep deployment_seasonality_profile editable per hospital and do not hardcode local months into the universal influenza pack.
- Preserve auditability: source → trigger → action → map should stay explicit.

9. Current workbook state reviewed

Reviewed workbook: HSIL (2).xlsx. The sheets Página20 and Página23 are blank legacy pages in the current file and do not contribute to the active logic. The abbreviations sheet acts as a glossary/reference sheet for human readers.

CodeBlue Workbook Architecture, Inclusion Crite \ria, and Search Strategy

Purpose and construction notes for the pitch deck and technical handoff

What this document does
- Explains what each workbook table is for.
- Explains how the tables were constructed and how information moves between them.
- States the inclusion, exclusion, and rerouting criteria used during evidence screening.
- Summarizes the search strategy used for evidence, policy, and deployment-context inputs.

Audience: pitch team, technical collaborator, and future maintainers

1. What the workbook is trying to do

The workbook is not a single literature table. It is a structured representation of how CodeBlue reasons about hospital-acquired influenza. The design deliberately separates patient acquisition risk, outbreak transmission context, post-influenza severity, secondary infections, staff-vector logic, molecular outbreak confirmation, policy/governance, and pathogen-pack configuration.

This separation matters because the project is not just predicting who gets influenza. It is trying to support reviewable hospital decisions in a dynamic setting: which ward is still worth protecting, where spread is emerging, which staff movement patterns are risky, and which interventions are allowed under policy.

The core design principle is: one layer estimates or stores evidence, a different layer translates that into triggers and actions, and a third layer adapts that logic to the pathogen and the local deployment context.

2. Design principles used to build the workbook

Principle	Why it exists
Causal-stage separation.	Acquisition risk, transmission context, severity after infection, secondary infections, surveillance logic, and policy logic were kept in different tables.
Anti-leakage rule.	Variables only known after influenza is established or after deterioration were not allowed into the acquisition table.
Time-aware logic.	Some features are invalid at admission but valid during later daily updates. The structure was designed to preserve that distinction.
One row = one factor or one rule.	Evidence rows do not represent entire papers. They represent extractable findings that can be reused computationally.
Human-reviewed actions only.	The policy layer proposes reviewable actions. It does not let the model autonomously order medication or irreversible interventions.
Universal pathogen logic + local deployment modifiers.	Influenza biology and CDC healthcare logic stay in the pathogen pack; local seasonality and burden are deployment inputs, not hardcoded pathogen truth.

3. Table map: what each table is for

Evidence layer

Table	Purpose
ha_influenza_risk_factors	Stores factors that help answer: who gets hospital-acquired influenza, or what reduces that risk. Only pre-infection or early-timepoint features belong here.

Table	Purpose
hospitalized_influenza_severity	Stores factors and burden findings that apply after influenza is already present, such as ICU admission, mortality, longer stay, or major clinical deterioration.
secondary_infections_in_influen	Stores findings on bacterial, fungal, resistant, or other secondary infections that occur after influenza.
pathogen_interactions	Stores which organisms appear in overlap/superinfection contexts and how those interactions matter.
transmission_context_support	Stores spread mechanics, room/unit overlap, clustering, hidden spread, burden of hospital dissemination, and non-genomic outbreak-support evidence.
staff_vector_transmission	Stores healthcare-worker-related spread logic such as presenteeism, asymptomatic burden, vaccination effects, role-specific transmission, and cross-coverage concerns.
advanced_outbreak_monitoring_to	Stores sequencing, WGS, Nanopore, or other molecular methods used to confirm or refine outbreaks.
library	Stores canonical feature names, aliases, and harmonization logic used across the workbook.

Policy and governance layer

Table	Purpose
policy_source_library	Stores the external authorities behind the logic: CDC influenza guidance, CDC antivirals, core IPC guidance, and local deployment sources.
policy_trigger_library	Stores operational events that should make the system pay attention. Triggers are not actions; they are the events that route actions.
action_library	Stores the stable catalog of reviewable actions. This table defines what each action is, who reviews it, what it targets, and what inputs it needs.
trigger_action_map	Stores the adaptive decision logic linking triggers to actions. This is where seasonality, vaccination, vulnerability, spread state, and capacity modify priorities.

Pathogen and deployment layer

Table	Purpose
influenza_pack_timing	Stores universal influenza timing assumptions such as incubation, contagious window, precaution duration, and prolonged-shedding modifiers.
influenza_pack_risk_features	Stores the influenza-specific prioritization of canonical features, especially host-vulnerability and vaccination features.
influenza_pack_interventions	Stores influenza-specific intervention hooks that connect the pathogen pack to the policy layer.
deployment_seasonality_profile	Stores editable local seasonality settings supplied by the hospital or deployment team, not hardcoded into the pathogen pack.

4. How the tables were constructed

The workbook was built from the top down, not by dumping papers into one spreadsheet. First, the target decision architecture was defined: evidence layer, policy layer, and pathogen/deployment layer. Then each literature finding or policy statement was routed into the table that best matched its causal stage and operational role.

Construction therefore followed three consecutive questions:

1. 1) What question is this paper or guidance actually answering?
2. 2) At what time in the hospital timeline is this information known?
3. 3) Is this an evidence row, a policy source, a trigger, an action, a pathogen modifier, or a deployment modifier?

That routing step is why one paper can generate rows in more than one table. For example, a paper may support both transmission-context rows and severity rows, but only if each extracted finding clearly fits the target table.

The action layer was intentionally split in two. `action_library` was kept stable and generic, while `trigger_action_map` carries the adaptive criteria. This prevents the action catalog from becoming a messy rules engine and makes the system easier to audit and extend.

5. Search strategy used

The searches were not built as one giant PubMed query. A multi-query approach was used because the workbook itself is split into evidence buckets. Each bucket had its own search logic so that acquisition, transmission, severity, secondary infections, staff-vector logic, and molecular outbreak monitoring did not get mixed.

Search bucket	Typical target papers	Primary destination table(s)
Acquisition risk	Who gets hospital-acquired influenza, who is protected, early/contextual vulnerability studies	ha_influenza_risk_factors
Severity and burden	Among infected patients, who deteriorates, dies, requires ICU, or has longer stay	hospitalized_influenza_severity
Secondary infection	Bacterial/fungal/resistant complications after influenza	secondary_infections_in_influen; pathogen_interactions
Transmission context	Same-room/same-unit spread, clustering, room structure, hidden spread, subtype-linked dissemination	transmission_context_support
Staff vector logic	HCW vaccination, presenteeism, asymptomatic burden, role-specific spread	staff_vector_transmission
Advanced monitoring	WGS, sequencing, Nanopore, genomic cluster confirmation	advanced_outbreak_monitoring_to
Policy/governance	CDC influenza healthcare guidance, antivirals, core IPC practices, local seasonality sources	policy_source_library; policy_trigger_library; action_library

The screening workflow was also staged:

- Title stage: each paper was mapped as Include, Other table, Exclude, or Borderline.
- Abstract stage: only papers that still matched the target table after reading the abstract were kept.
- Full-text stage: row extraction only occurred when the finding was clearly extractable and compatible with the existing sheet structure.
- Deduplication: duplicated uploads and duplicate reference-list entries were collapsed. If both a preprint and published paper existed, the published paper was preferred.

6. Inclusion, exclusion, and rerouting criteria

The project used a strict routing philosophy. A paper was not included just because it was about influenza in hospitals. It had to answer the right table question.

ha_influenza_risk_factors

Include:

- Baseline vulnerability known before infection

- Protective factors such as vaccination
- Early structural or rooming context
- Early clinical/radiology/lab context available before hospital-acquired influenza is established

Exclude:

- ICU admission, mortality, complications, or outcomes clearly downstream of infection
- Hospital-acquired influenza status as if it were a predictor
- Secondary bacterial/fungal infection after influenza

Reroute to: Severity, secondary-infection, or transmission tables depending on what the paper actually measures

hospitalized_influenza_severity

Include:

- ICU use
- Mechanical ventilation
- Mortality
- Longer stay or clinical deterioration after influenza is present
- Comparisons between hospital-acquired and community-acquired influenza outcomes

Exclude:

- Pre-infection vulnerability used to predict acquisition
- Pure room-exposure or outbreak-spread studies

Reroute to: Acquisition or transmission tables

secondary_infections_in_influen

Include:

- Bacterial superinfection
- Sepsis after influenza
- Hospital-acquired resistant-pathogen complications after influenza

Exclude:

- Pure acquisition studies
- Pure virology or non-complication transmission studies

Reroute to: Pathogen interactions if the main value is the organism profile

pathogen_interactions

Include:

- Specific organisms in overlap or superinfection
- Resistance profiles
- Which pathogens matter in mixed post-influenza contexts

Exclude:

- Pure influenza acquisition risk or pure mortality comparisons

Reroute to: Secondary infections when the question is complication risk rather than organism identity

transmission_context_support

Include:

- Same-room or same-unit exposure
- Room layout, ventilation, double-room structure
- Clustering, hidden spread, importation pressure, subtype-linked dissemination
- Pragmatic prevention strategies that change spread

Exclude:

- Post-influenza severity without spread logic
- Pure genomic confirmation where the value is only molecular linkage

Reroute to: Advanced outbreak monitoring when sequencing is the main contribution

staff_vector_transmission

Include:

- HCW vaccination effects
- Presenteeism
- Asymptomatic or paucisymptomatic HCW burden
- Role-specific transmission networks
- Cross-coverage or staff-source control context

Exclude:

- Papers about general staff knowledge or attitudes with no operational transmission value

Reroute to: Transmission context when the paper is mainly about ward-level spread rather than staff vector specifics

advanced_outbreak_monitoring_to

Include:

- WGS
- Nanopore
- Molecular outbreak linkage
- Sequencing used to confirm or refine transmission chains

Exclude:

- Routine non-genomic transmission context papers
- Policy/guideline papers

Reroute to: Transmission context or policy layer depending on the paper

7. Why the policy layer was built this way

The policy layer was intentionally separated into four tables because each answers a different question.

Table	Question it answers	Why it is separate
policy_source_library	What external or local source authorizes this logic?	Keeps CDC, national, and state guidance explicit and auditable.
policy_trigger_library	What event should make the system pay attention?	Prevents actions from being mixed with events.
action_library	What reviewable action exists?	Keeps a stable verb catalog without burying adaptive rules inside it.
trigger_action_map	When should this action be activated, escalated, downgraded, or suppressed?	This is where the adaptive criteria live: seasonality, vaccination, vulnerability, spread state, and capacity.

8. Why the pathogen pack and local seasonality were separated

The influenza pathogen pack was made universal on purpose. It stores influenza timing, high-risk features, and intervention hooks that should remain true across deployments. Local seasonality was not hardcoded into that pack.

Instead, local seasonality was moved into `deployment_seasonality_profile`. The reason is practical: when another hospital adopts CodeBlue, the team should be able to change the local seasonality profile without rewriting the influenza pack.

This produces a cleaner architecture:

- • Influenza pack = universal pathogen logic
- • Deployment seasonality profile = local operational expectation
- • Current epidemiology status = live override when the real world diverges from the usual seasonal profile

9. Pitch-ready explanation

Short version for the pitch:

We did not build a single spreadsheet of papers. We built a computational knowledge structure. One layer stores evidence about who acquires hospital-acquired influenza, how it spreads, how severe it becomes, and how staff contribute. A second layer converts that knowledge into reviewable triggers and actions under CDC-backed policy logic. A third layer keeps influenza-specific assumptions separate from local deployment settings such as seasonality. That separation is what makes CodeBlue reusable, auditable, and adaptable across hospitals.

Technical version for the handoff:

The workbook is a layered architecture. Evidence tables are causal-stage specific. Policy tables separate authorities, triggers, actions, and adaptive mappings. The influenza pack is universal, while

seasonality and current epidemiology are deployment inputs. Search and screening used multi-query routing, strict inclusion criteria, and rerouting to avoid leakage and category collapse.

10. Practical notes for the technical collaborator

- Do not merge the evidence tables into one monolithic table.
- Do not move post-infection outcomes into `ha_influenza_risk_factors`.
- Treat `action_library` as a stable catalog, not a rules engine.
- Keep adaptive logic in `trigger_action_map`.
- Keep local seasonality editable at the deployment level.
- Prefer published papers over preprints when both are present.
- Use `ready_partial` when evidence is mixed, weaker, ecological, or only indirectly aligned.
- Allow one paper to populate multiple tables only when each extracted finding clearly fits its destination.

11. Current gaps and likely next additions

The current workbook is already coherent enough for a pitch and for continued technical work. The most likely next additions are a deployment threshold table, a broader hospital deployment profile, and a governance/audit log for overrides. These are enhancements, not prerequisites for understanding the current architecture.